Darya Zare

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Education

- **University of Zurich (UZH)** *Zurich, Switzerland* Sep. 2024 – Present
  - Master of Science in Computer Science; Major in Artificial Intelligence, Minor in Data Science
  - Special student at ETH Zurich
  - Coursework: Large Language Models; Probabilistic AI; Advanced Topics in AI; Advanced Techniques of Machine Translation; Machine Learning for NLP I & II; AI in Education; Deep Learning; Foundations of Data Science; Human–Computer Interaction in AI; Network Science
- **Amirkabir University of Technology** *Tehran, Iran* Sep. 2018 – Sep. 2023
  - Bachelor of Science in Computer Engineering

Research Experience

- **ETH AI Center, ETH Zurich** *Student Researcher – AI Safety* Jun. 2026 – Present
  - Investigating how anthropomorphic and emergent misalignment arise in language models following fine-tuning, with a focus on comparing base and fine-tuned models.
  - Developing multi-agent evaluation settings to study how dimensions of Anthropomorphic Misalignment Risk (AMR) emerge and evolve over long-horizon interaction trajectories.
- **ETH Zurich** *Research Assistant – Prof. Elliott Ash* Apr. 2025 – Present
  - Developed NLP methods for compliance analysis of medical device regulatory documents, focusing on long-document evidence retrieval.
  - Co-developed a RAG-based regulatory review system that grounds answers in retrieved document evidence.
  - Developed an uncertainty-aware knowledge distillation approach that transfers soft relevance judgments from a teacher LLM to a compact student model, producing calibrated relevance scores for filtering misleading retrieved evidence in RAG systems.
- **University of Zurich (UZH)** *Research Assistant – Prof. Thomas Kurer* Dec. 2024 – Jan. 2025
  - Applied NLP methods to French political survey data to analyze government policy effects.
  - Worked with open-source LLMs, including LLaMA and Qwen, using prompt-tuning strategies.
  - Evaluated model outputs for accuracy and robustness.

Industry Experience

- **Metodata** *Data Scientist* Jun. 2021 – Aug. 2024
  - Worked on social media data analysis for Persian and English text corpora.
  - Designed and evaluated machine learning pipelines for NLP tasks.
  - Deployed trained machine learning models to production servers.

Selected Publications

- **Exploring Relationships Between Cryptocurrency News Outlets and Influencers** Feb. 2024 – Mar. 2024
  - Performed prompt tuning to generate buy/sell signals from social media data.
  - Conducted cross-correlation analysis between news outlets and influencer signals.
- **Comparing Methods for Creating a National Random Sample of Twitter Users** Jan. 2023 – Jul. 2023
  - Developed Twitter user sampling pipelines using social-network, profile, and activity-based signals.
  - Engineered user-level features and evaluated demographic bias across sampling strategies.
  - Applied multilevel regression and poststratification to assess state-level and national representativeness.

Selected Projects

- **Hybrid Listing Search and Relevance Ranking for Robin Real** *Datathon* April 2026
  - Built a retrieval and ranking system for real-estate listings that interprets natural language queries.
  - Contributed to a team-based 24-hour prototype that won **1st place** in the main challenge; team also received 3rd place in the Anthropic side challenge.
- **Network Certification & Robustness (RTAI)** *ETH Zurich* Fall 2025
  - Certified neural encoder/decoder properties using MILP with Gurobi and DeepPoly-based abstract interpretation.
  - Applied PGD-based adversarial training.
- **Movie Recommendation Chatbot** *UZH* Fall 2025
  - Built a conversational movie recommendation chatbot that combined natural language understanding, embedding-based representations, and knowledge graph reasoning to answer factual movie-related queries.
  - Generated personalized recommendations by modeling both collaborative signals from similar users and individual user preferences based on favorite movie features.
  - Designed a dialogue pipeline to extract user preferences, intents, and entity-level information, enabling recommendation, retrieval, and question-answering workflows.
- **Airline Network Community Detection** *UZH* Fall 2024
  - Analyzed the global airline network using graph-based methods to uncover community structure and characterize network topology at scale.
  - Implemented Louvain and Clique Percolation Method (CPM) algorithms to detect non-overlapping and overlapping airport communities.

Skills

- **Programming Languages:** *Python, R, C, C++, Java, JavaScript, HTML/CSS*
- **ML / NLP:** *PyTorch, TensorFlow, Transformers, Scikit-learn, Gensim*
- **Tools:** *Pandas, NumPy, OpenCV, Git, SQL, Docker, Streamlit*
- **Operating Systems:** *Linux, macOS, Windows*